GMI German-Malaysian Institute

DIPLOMA PROGRAMME

REVISION EXERCISE

RSA FOR MESSAGE

Message Encryption and Decryption (Confidentiality)

Academic Period : July 2026

Subject : CRYPTOGRAPHY

Code : CBS 2373

Type : REVISION QUESTION SET (STUDENT COPY) Number of Questions : 5

Tool : MICROSOFT EXCEL

Suggested Time : 2 HOURS (SELF-PACED)

GENERAL INSTRUCTIONS:

1. Write down clearly (with capital letters) your name, group and the date at the bottom of this page.
2. Attempt every question using Microsoft Excel. Do not look at the answer file until you have finished.
3. Use one worksheet for each question (QUESTION 1 to QUESTION 5) and show the MOD calculation in the cells.
4. Fill in the answer table and paste a screenshot of your Excel result under each question.
5. Compare your work with the separate answer file and tick the self-check box.

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Semester/Trade/Group	**:**	**SEMESTER 5 / DCBS / CBS 1**
Date	**:**	**9 / 10 / 2026**

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No	Revision Task	Questions	CLO	PO	Level of Difficulty
1	To generate key for RSA public key and private key.	Q1, Q2, Q3	2	4	P3
2	To encrypt a message using the RSA public key.	Q2, Q4, Q5	2	4	P3
3	To decrypt a ciphertext using the RSA private key.	Q2, Q3, Q4, Q5	2	4	P3
4	To compare encryption by public key and by private key.	Q5	2	4	P3

Learning Objectives:

1. To generate key for RSA public key and private key.
2. To encrypt a message using the RSA public key.
3. To decrypt a ciphertext using the RSA private key.
4. To compare encryption by public key and by private key.

These revision tasks are based on RSA key generation, encryption with the public key and decryption with the private key. All numbers are small so they can be checked with a calculator.

You are required to answer in Microsoft Excel (one worksheet per question) and keep your Excel file with the MOD calculation. All answers are based on your Excel solution.

Excel Guide

What to calculate	**Excel formula / method**
n and φ(n)	=p*q and =(p-1)*(q-1)
Check e	=GCD(e,phi). The answer must be 1.
Find d	List d = 1, 2, 3 ... φ(n)−1. In the next column use =MOD(e*d,phi). Pick the d that gives 1.
Small power mod n	=MOD(POWER(M,e),n), for example =MOD(POWER(11,5),65)
Large exponent (e.g. d)	POWER becomes too large for Excel. Use a repeated-multiplication table: first cell =MOD(M,n); next cell =MOD(previous*M,n); copy down until the row number equals the exponent.
Letter to number	A = 1 … Z = 26

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Tasks	Questions	Self check
To generate key for RSA public key and private key.	**QUESTION 1:** Bob wants to receive secret messages using RSA. Generate Bob's RSA public key and private key based on the prime numbers given by the lecturer. Rule for e: choose the **smallest** integer e where 1 < e < φ(n) and GCD(e, φ(n)) = 1. Use an Excel table to find d where MOD(e × d, φ(n)) = 1. **Given by Lecturer Value** **Prime number p** 7 **Prime number q** 11 **Item Answer** **n** **φ(n)** **e** **Public key (e, n)** **d** **Private key (d, n)** **Screenshot the Excel result:**	☐

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QUESTION 1: to generate key of RSA (public key and private key) based on the prime numbers given by Lecturer. Choose the smallest e.

	Given by Lecturer
p	7
q	11

Bedul CBS 1

ANSWER:

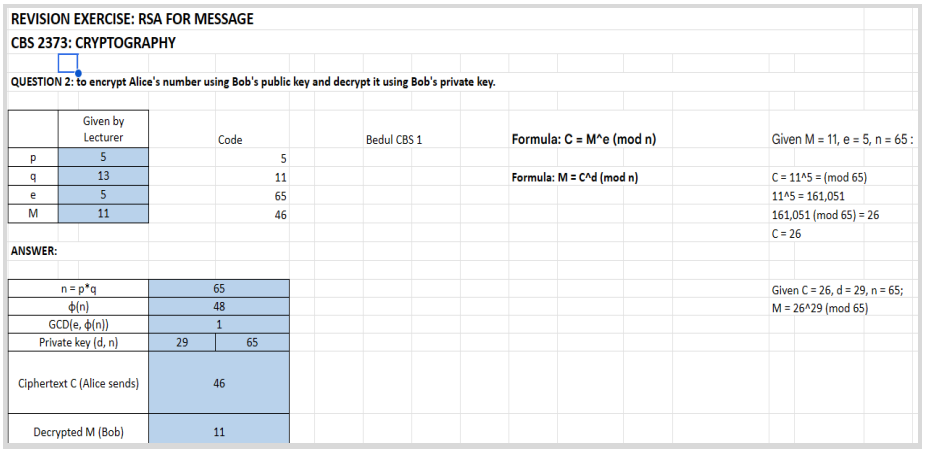
n = p*q	77
ϕ(n) = (p-1)*(q-1)	60
e	7
Public key (e, n)	7, 77
d	43
Private key (d, n)	43,77

PRIME NUMBER (1-100)				
2	3	5	7	11
13	17	19	23	29
31	37	41	43	47
53	59	61	67	71
73	79	83	89	97

1	<	E	<	phi (n)					1	<	D	<	phi (n)
		E	GCD(E, phi(n))		gcd (e, phi(n)) = 1		Cofactor phi (n) = 60		D	e * D	d * e mod phi (n) = 1		
		2	2		gcd (2, 60) = 1				2	14	14		
		3	3				60/2 = 30		3	21	21		
		4	4			60/1 = 60	30/2 = 15		4	28	28		
		5	5			60/2 = 30	15/3 =5		5	35	35		
		6	6			60/3 = 20	5/5 = 1		6	42	42		
		7	1	Choose as "e"		60/4 = 15			7	49	49		
		8	4			60/5 = 12	60 = 2x2x3x5		8	56	56		
		9	3			60/6 = 10	e = 7,11,13,17,19		9	63	3		
		10	10			60/7 = ?			10	70	10		
		11	1						11	77	17		
		12	12						12	84	24		
		13	1						13	91	31		
		14	2						14	98	38		
		15	15						15	105	45		
		16	4						16	112	52		
		17	1						17	119	59		
		18	6						18	126	6		
		19	1						19	133	13		
		20	20						20	140	20		
		21	3						21	147	27		
		22	2						22	154	34		
		23	1						23	161	41		
		24	12						24	168	48		
		25	5						25	175	55		
		26	2						26	182	2		
		27	3						27	189	9		
		28	4						28	196	16		
		29	1						29	203	23		
		30	30						30	210	30		
		31	1						31	217	37		
		32	4						32	224	44		
		33	3						33	231	51		
		34	2						34	238	58		
		35	5						35	245	5		
		36	12						36	252	12		
		37	1						37	259	19		
		38	2						38	266	26		
		39	3						39	273	33		
		40	20						40	280	40		
		41	1						41	287	47		
		42	6						42	294	54		
		43	1						43	301	1		Choose as "d"
		44	4						44	308	8		

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Tasks	Questions	Self check
To encrypt with the public key and decrypt with the private key.	**QUESTION 2:** Alice wants to send a secret number to Bob. Generate Bob's private key, then encrypt Alice's number using Bob's **public key**. Finally, show that Bob can decrypt the ciphertext using his **private key**. Use C = M^e mod n and M = C^d mod n. **Given by Lecturer Value** **Prime number p** 5 **Prime number q** 13 **Public exponent e** 5 **Alice's plaintext M** 11 **Item Answer** **n** **φ(n)** **GCD(e, φ(n))** **Private key (d, n)** **Ciphertext C (Alice sends)** **Decrypted M (Bob)** **Screenshot the Excel result:** 	☐

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QUESTION 2: to encrypt Alice's number using Bob's public key and decrypt it using Bob's private key.

	Given by Lecturer	Code	Bedul CBS 1
p	5	5	
q	13	11	
e	5	65	
M	11	46	

ANSWER:

n = p*q	65	
ϕ(n)	48	
GCD(e, ϕ(n))	1	
Private key (d, n)	29	65
Ciphertext C (Alice sends)	46	
Decrypted M (Bob)	11	

Formula: C = M^e (mod n)
Given M = 11, e = 5, n = 65 :

Formula: M = C^d (mod n)
C = 11^5 = (mod 65)
11^5 = 161,051
161,051 (mod 65) = 26
C = 26

Given C = 26, d = 29, n = 65;
M = 26^29 (mod 65)

1 <	D	< phi (n)
	D	e * D	MOD(e*D, phi(n))
	2	10	10
	3	15	15
	4	20	20
	5	25	25
	6	30	30
	7	35	35
	8	40	40
	9	45	45
	10	50	2
	11	55	7
	12	60	12
	13	65	17
	14	70	22
	15	75	27
	16	80	32
	17	85	37
	18	90	42
	19	95	47
	20	100	4
	21	105	9
	22	110	14
	23	115	19
	24	120	24
	25	125	29
	26	130	34
	27	135	39
	28	140	44
	29	145	1
	30	150	6
	31	155	11
	32	160	16
	33	165	21
	34	170	26
	35	175	31
	36	180	36
	37	185	41

k	C ^ k mod n
k (expone	C^k mod n
1	46
2	36
3	31
4	61
5	11
6	51
7	6
8	16
9	21
10	56
11	41
12	1
13	46
14	36
15	31
16	61
17	11
18	51
19	6
20	16
21	21
22	56
23	41
24	1
25	46
26	36
27	31
28	61
29	11	Plaintext Value Match with M

1 <	E	< ph	
	E	GCD(E, phi(n))	
	2	2	
	3	3	
	4	4	
	5	1	Choose as "e"
	6	6	
	7	1	
	8	8	
	9	3	
	10	2	
	11	1	
	12	12	
	13	1	
	14	2	
	15	3	
	16	16	
	17	1	
	18	6	
	19	1	
	20	4	
	21	3	
	22	2	
	23	1	
	24	24	
	25	1	
	26	2	
	27	3	
	28	4	
	29	1	
	30	6	
	31	1	
	32	16	
	33	3	
	34	2	
	35	1	
	36	12	
	37	1	

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Tasks	Questions	Self check
To decrypt a ciphertext with the private key.	**QUESTION 3:** Bob has published his public exponent and keeps his primes secret. Bob receives a ciphertext from Alice. Find Bob's private key, decrypt the ciphertext, then check your answer by encrypting the plaintext again with the public key. **Given by Lecturer Value** **Prime number p** 5 **Prime number q** 17 **Public exponent e** 3 **Ciphertext received C** 24 **Item Answer** **n** **φ(n)** **Public key (e, n)** **Private key (d, n)** **Plaintext M** **Check: M^e mod n** **Screenshot the Excel result:**	☐

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QUESTION 3: to find Bob's private key, decrypt the ciphertext received and check the answer by encrypting the plaintext again.

	Given by Lecturer
p	5
q	17
e	3
C	24

Bedul CBS 1

d = 43

ANSWER:

n	85		
ϕ(n)	64		
Public key (e, n)	3	85	255
Private key (d, n)	43	85	3655
Plaintext M	14		
Check: M^e mod n	24		

1 <	D	<	phi (n)			k	C ^ k mod n
	D	e * D	MOD(e*D, phi(n))			k (expone	C^k mod n
	2	6	6			1	24
	3	9	9			2	66
	4	12	12			3	54
	5	15	15			4	21
	6	18	18			5	79
	7	21	21			6	26
	8	24	24			7	29
	9	27	27			8	16
	10	30	30			9	44
	11	33	33			10	36
	12	36	36			11	14
	13	39	39			12	81
	14	42	42			13	74
	15	45	45			14	76
	16	48	48			15	39
	17	51	51			16	1
	18	54	54			17	24
	19	57	57			18	66
	20	60	60			19	54
	21	63	63			20	21
	22	66	2			21	79
	23	69	5			22	26
	24	72	8			23	29
	25	75	11			24	16
	26	78	14			25	44
	27	81	17			26	36
	28	84	20			27	14
	29	87	23			28	81
	30	90	26			29	74
	31	93	29			30	76
	32	96	32			31	39
	33	99	35			32	1
	34	102	38			33	24
	35	105	41			34	66
	36	108	44			35	54
	37	111	47			36	21
	38	114	50			37	79
	39	117	53			38	26
	40	120	56			39	29
	41	123	59			40	16
	42	126	62			41	44
	43	129	1	Choose as "d"		42	36
	44	132	4			43	14
	45	135	7				

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Tasks	Questions	Self check
To encrypt and decrypt a text message letter by letter.	**QUESTION 4:** Alice sends the word **EXCEL** to Bob, one letter at a time. Convert each letter to a number using A = 1, B = 2, ... Z = 26. Encrypt each number with Bob's public key, then let Bob decrypt every ciphertext with his private key.	☐

Given by Lecturer Value

Prime number p 5

Prime number q 11

Public exponent e 3

Message EXCEL

Item Answer

n
φ(n)
Private key (d, n)

Letter	M (number)	C = M^e mod n	Bob decrypts: M = C^d mod n	Letter back
E				
X				
C				
E				
L				

Screenshot the Excel result:

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QUESTION 4: to encrypt the word EXCEL letter by letter (A = 1 ... Z = 26) using Bob's public key and decrypt every ciphertext with Bob's private key.

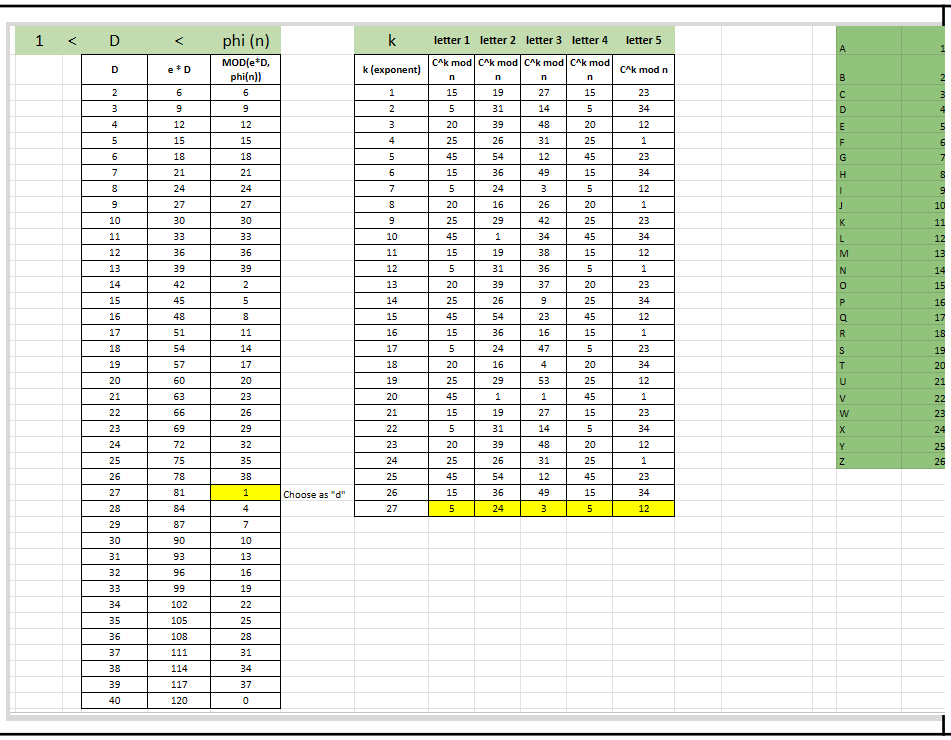
	Given by Lecturer
p	5
q	11
e	3
Message	EXCEL

Bedul CBS1

ANSWER:

n = p * q	55	
phi (n) = (p-1)*(q-1)	40	
private key (d, n)	27	55

Letter	M (number)	C = M^e mod n	Bob decrypts: C^d mod n	Letter back
E	5	15	5	E
X	24	19	24	X
C	3	27	3	C
E	5	15	5	E
L	12	23	12	L
Ciphertext sent	EXCEL			



1	<	D	<	phi (n)

D	e * D	MOD(e*D, phi(n))	
2	6	6	
3	9	9	
4	12	12	
5	15	15	
6	18	18	
7	21	21	
8	24	24	
9	27	27	
10	30	30	
11	33	33	
12	36	36	
13	39	39	
14	42	2	
15	45	5	
16	48	8	
17	51	11	
18	54	14	
19	57	17	
20	60	20	
21	63	23	
22	66	26	
23	69	29	
24	72	32	
25	75	35	
26	78	38	
27	81	1	Choose as "d"
28	84	4	
29	87	7	
30	90	10	
31	93	13	
32	96	16	
33	99	19	
34	102	22	
35	105	25	
36	108	28	
37	111	31	
38	114	34	
39	117	37	
40	120	0	

k	letter 1	letter 2	letter 3	letter 4	letter 5
k (exponent)	C^k mod n	C^k mod n	C^k mod n	C^k mod n	C^k mod n
1	15	19	27	15	23
2	5	31	14	5	34
3	20	39	48	20	12
4	25	26	31	25	1
5	45	54	12	45	23
6	15	36	49	15	34
7	5	24	3	5	12
8	20	16	26	20	1
9	25	29	42	25	23
10	45	1	34	45	34
11	15	19	38	15	12
12	5	31	36	5	1
13	20	39	37	20	23
14	25	26	9	25	34
15	45	54	23	45	12
16	15	36	16	15	1
17	5	24	47	5	23
18	20	16	4	20	34
19	25	29	53	25	12
20	45	1	1	45	1
21	15	19	27	15	23
22	5	31	14	5	34
23	20	39	48	20	12
24	25	26	31	25	1
25	45	54	12	45	23
26	15	36	49	15	34
27	5	24	3	5	12

A	1
B	2
C	3
D	4
E	5
F	6
G	7
H	8
I	9
J	10
K	11
L	12
M	13
N	14
O	15
P	16
Q	17
R	18
S	19
T	20
U	21
V	22
W	23
X	24
Y	25
Z	26

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Tasks	Questions	Self check
To compare public-key and private-key encryption.	**QUESTION 5:** Compare **encryption with the public key** and **encryption with the private key**. Use the same plaintext for both methods and complete the table. **Given by Lecturer Value** **Prime number p** 7 **Prime number q** 17 **Public exponent e** 5 **Plaintext M** 20 **Item Answer** **n** **φ(n)** **Private key (d, n)** **Step Answer** (a) Encrypt M with the PUBLIC key: C = M^e mod n (b) Decrypt C with the PRIVATE key: C^d mod n (c) Which method gives confidentiality (only Bob can read)? **Screenshot the Excel result:**	☐

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QUESTION 5: to encrypt Alice's number using the public key and decrypt it using the private key (confidentiality).

	Given by Lecturer
p	7
q	17
e	5
M	20

Bedul CBS1

ANSWER:

n	119	
ϕ(n)	96	
Private key (d, n)	77	119
A) Encrypt M with the public key: C = M^e mod n	90	
B) Decrypt C with the private key: C^d mod n	20	
C) Which method gives confidentiality (Only bob can read?)	Encryption with the PUBLIC key	

1 <	D	<	phi (n)			k	C ^ k mod n
	D	e * D	MOD(e*D, phi(n))			k (expone	C^k mod n
	2	10	10			1	90
	3	15	15			2	8
	4	20	20			3	6
	5	25	25			4	64
	6	30	30			5	48
	7	35	35			6	36
	8	40	40			7	27
	9	45	45			8	50
	10	50	50			9	97
	11	55	55			10	43
	12	60	60			11	62
	13	65	65			12	106
	14	70	70			13	20
	15	75	75			14	15
	16	80	80			15	41
	17	85	85			16	1
	18	90	90			17	90
	19	95	95			18	8
	20	100	4			19	6
	21	105	9			20	64
	22	110	14			21	48
	23	115	19			22	36
	24	120	24			23	27
	25	125	29			24	50
	26	130	34			25	97
	27	135	39			26	43
	28	140	44			27	62
	29	145	49			28	106
	30	150	54			29	20
	31	155	59			30	15
	32	160	64			31	41
	33	165	69			32	1
	34	170	74			33	90
	35	175	79			34	8
	36	180	84			35	6
	37	185	89			36	64
	38	190	94			37	48
	39	195	3			38	36
	40	200	8			39	27
	41	205	13			40	50
	42	210	18			41	97
	43	215	23			42	43
	44	220	28			43	62
	45	225	33			44	106

38	190	94				37	48
39	195	3				38	36
40	200	8				39	27
41	205	13				40	50
42	210	18				41	97
43	215	23				42	43
44	220	28				43	62
45	225	33				44	106
46	230	38				45	20
47	235	43				46	15
48	240	48				47	41
49	245	53				48	1
50	250	58				49	90
51	255	63				50	8
52	260	68				51	6
53	265	73				52	64
54	270	78				53	48
55	275	83				54	36
56	280	88				55	27
57	285	93				56	50
58	290	2				57	97
59	295	7				58	43
60	300	12				59	62
61	305	17				60	106
62	310	22				61	20
63	315	27				62	15
64	320	32				63	41
65	325	37				64	1
66	330	42				65	90
67	335	47				66	8
68	340	52				67	6
69	345	57				68	64
70	350	62				69	48
71	355	67				70	36
72	360	72				71	27
73	365	77				72	50
74	370	82				73	97
75	375	87				74	43
76	380	92				75	62
77	385	1	Choose as "d"			76	106
78	390	6				77	20
79	395	11					
80	400	16					
81	405	21					
82	410	26					
83	415	31					
84	420	36					
85	425	41					
86	430	46					
87	435	51					
88	440	56					
89	445	61					
90	450	66					
91	455	71					
92	460	76					
93	465	81					
94	470	86					

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